

Supplementary Material

1 Electron equivalents of organic substrates

Table 1. Electron equivalents of organic substrates and degradation intermediates and end products (Zhao et al., 2020b) (Rittmann & McCarty, 2020).

Molecules	Degradation reaction	e ⁻ eq per carbon
Glucose	$C_6H_{12}O_6 + 6H_2O \rightarrow 6CO_2 + 24H^+ + 24e^-$	$24/6 = 4$
Propionate	$C_3H_6O_2 + 4H_2O \rightarrow 3CO_2 + 14H^+ + 14e^-$	$14/3 = 4.67$
Ethanol	$C_2H_6O + 3H_2O \rightarrow 2CO_2 + 12H^+ + 12e^-$	$12/2 = 6$
Acetate	$C_2H_4O_2 + 2H_2O \rightarrow 2CO_2 + 8H^+ + 8e^-$	$8/2 = 4$
Formate	$CH_2O_2 \rightarrow CO_2 + 2H^+ + 2e^-$	$2/1 = 2$
Butyrate	$C_4H_8O_2 + 6H_2O \rightarrow 4CO_2 + 20H^+ + 20e^-$	$20/4 = 4$
Valerate	$C_5H_9O_2 + 8H_2O \rightarrow 5CO_2 + 25H^+ + 25e^-$	$25/5 = 5$

Table 2. Distribution of electron equivalents (%) for the MFCs at each sample point.

	Time of sample (h)	Average electron balance (%)							Current	Total	Unknown
		Acetic acid	Formic acid	Propionic acid	Butyric acid	Isobutyric acid	Valeric acid	Isovaleric acid			
Acetate - stirred	0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	100.0	0.0
	1	96.3	0.6	1.4	0.0	0.0	0.0	0.0	0.2	98.5	1.5
	2.5	95.6	0.5	4.1	0.0	0.0	0.0	0.0	1.1	101.3	0
	3.5	92.0	0.5	1.6	0.0	0.0	0.0	0.0	1.8	95.8	4.2
	4.5	88.7	0.5	1.7	0.02	0.0	0.0	0.0	2.6	93.4	6.6
	24	39.4	0.4	3.1	0.7	0.2	0.0	0.0	15.8	59.6	40.4
	26	32.2	0.4	2.1	0.0	0.2	0.0	0.0	17.2	52.3	47.7
	28	25.8	0.6	1.9	0.0	0.2	0.0	0.0	18.7	47.2	52.8
	48	4.4	0.2	0.9	0.0	1.7	0.0	0.0	29.6	36.8	63.2
	51	2.7	0.4	0.4	0.0	1.7	0.0	0.0	31.7	36.9	63.1
	71	4.6	0.4	1.1	0.0	1.7	0.0	0.0	37.8	45.6	54.4
	96	5.3	0.4	1.4	0.0	1.0	0.0	0.0	45.7	53.6	46.4
	160	5.8	0.3	2.2	0.0	3.8	0.0	0.0	63.1	75.2	24.8
Acetate – non-stirred	0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	100.0	0.0
	1	98.2	0.5	1.4	0.0	0.0	0.0	0.0	0.1	100.3	0.0
	2.5	97.6	0.5	3.3	0.0	0.0	0.0	0.0	1.0	102.3	0.0
	3.5	96.1	0.4	2.1	0.0	0.0	0.0	0.0	1.7	100.3	0.0
	4.5	95.9	0.4	2.6	0.0	0.0	0.0	0.0	2.7	101.5	0.0
	24	63.1	1.2	4.1	0.0	0.2	0.0	0.0	19.8	88.3	11.7
	26	55.1	1.0	3.8	0.4	0.2	0.0	0.0	21.5	82.0	18.0
	28	50.6	0.5	3.2	0.1	0.2	0.0	0.0	23.3	77.9	22.1
	48	26.9	1.1	2.4	0.0	1.4	0.0	0.0	38.8	70.5	29.5
	51	25.6	2.4	5.2	0.0	1.6	0.0	0.0	42.0	76.7	23.3
	71	13.3	1.1	1.8	0.0	1.6	0.0	0.0	57.6	75.4	24.6
	96	10.9	0.9	1.2	0.0	1.1	0.0	0.0	73.0	87.1	12.9
	160	9.8	0.7	2.3	0.0	3.7	0.0	0.0	105.3	121.8	0.0

Table 2. continued.

	Time of sample (h)	Average electron balance (%)							Current	Total	Unknown
		Acetic acid	Formic acid	Propionic acid	Butyric acid	Isobutyric acid	Valeric acid	Isovaleric acid			
Acetate - stirred	0	2.71	0.55	1.33	0.00	1.64	0.00	0.00	0.00	100.0	0.0
	1	12.18	0.24	7.06	1.15	8.44	0.45	0.45	0.27	30.2	69.8
	2.5	13.44	0.32	6.52	1.09	8.74	0.66	0.66	0.63	32.1	67.9
	3.5	15.54	0.10	8.25	1.02	9.20	0.63	0.63	1.30	36.7	63.3
	4.5	14.35	0.01	7.22	1.10	11.35	0.66	0.66	1.75	37.1	62.9
	24	31.28	0.88	11.16	2.64	17.39	0.48	0.48	9.13	73.4	26.6
	26	31.73	0.86	11.91	2.23	15.93	0.33	0.33	9.81	73.1	26.9
	28	32.13	0.82	12.60	2.13	15.70	0.31	0.31	10.44	74.4	25.6
	48	32.65	0.82	20.24	2.52	18.22	0.30	0.30	15.73	90.8	9.2
	51	27.71	1.05	23.03	3.39	18.59	0.37	0.37	18.85	93.4	6.6
	71	22.78	1.28	25.82	4.25	18.96	0.49	0.49	21.96	96.0	4.0
	96	10.67	0.36	22.40	3.51	24.95	0.67	0.67	28.34	91.6	8.4
	160	7.58	0.34	20.98	2.55	18.71	0.88	0.88	46.66	98.6	1.4
Acetate – non-stirred	0	11.09	2.90	5.85	0.00	4.12	0.00	0.00	0.00	100.0	0.0
	1	11.04	1.53	3.14	0.00	5.19	0.00	0.00	0.25	21.2	78.8
	2.5	13.24	1.72	4.70	0.00	10.85	0.00	0.00	0.81	31.3	68.7
	3.5	14.69	1.93	4.79	0.00	10.30	0.00	0.00	1.69	33.4	66.6
	4.5	11.47	0.30	2.66	0.00	7.89	0.00	0.00	2.29	24.6	75.4
	24	34.36	1.19	10.75	0.00	12.95	0.00	0.00	11.77	71.0	29.0
	26	36.74	0.23	12.77	0.00	13.47	0.00	0.00	12.66	75.9	24.1
	28	36.09	0.29	12.98	0.00	14.54	0.00	0.00	13.64	77.5	22.5
	48	19.32	0.65	12.09	0.36	12.52	0.00	0.00	21.97	66.9	33.1
	51	14.63	0.51	10.57	0.11	12.02	0.00	0.00	26.69	64.5	35.5
	71	9.94	0.38	9.05	0.31	11.52	0.00	0.00	31.40	62.6	37.4
	96	9.43	0.75	4.18	0.00	9.31	0.00	0.00	41.31	65.0	35.0
	160	8.30	0.72	2.12	0.00	9.01	0.00	0.00	62.79	82.9	17.1

2 Calculation of the rates of the different degradation pathways

The linear plots below were used to obtain the kinetic rate constant k :

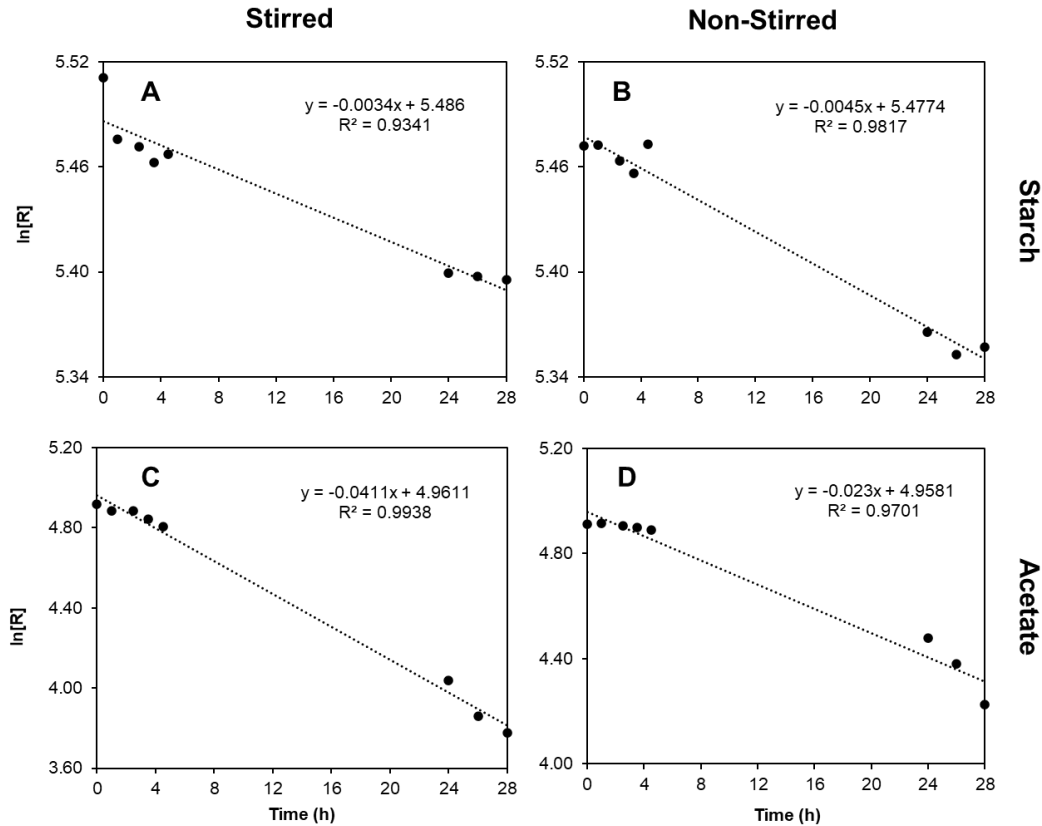


Figure 1. Linear plots of time vs the natural logarithm of the reactant concentration for: (A) starch-fed MFCs under stirred conditions; (B) starch-fed MFCs under non-stirred conditions; (C) acetate-fed MFCs under stirred conditions; and (D) acetate-fed MFCs under non-stirred conditions.

Rates were calculated by constructing a linear plot of the time (first 28 h) vs the natural logarithm ($\ln$) of the reactant concentration (Figure 1) based on Equation below.

The hydrolysis-fermentation pathway (conversion of starch to acetate) was calculated as follows:

Stirred conditions:

$$\ln(\text{starch}) = -0.0034t + 5.486$$

$$k = 0.0034 \text{ h}^{-1}$$

Non-stirred conditions:

$$\ln(\text{starch}) = -0.0045t + 5.4774$$

$$k = 0.0045 \text{ h}^{-1}$$

Rates of acetate consumption/electrogenesis was calculated as follows:

Stirred conditions:

$$\ln(\text{acetate}) = -0.0411t + 4.9611$$

$$k = 0.0411 \text{ h}^{-1}$$

Non-stirred conditions:

$$\ln(\text{acetate}) = -0.0232t + 4.9581$$

$$k = 0.0232 \text{ h}^{-1}$$

3 Total VFA production/consumption for individual batch runs

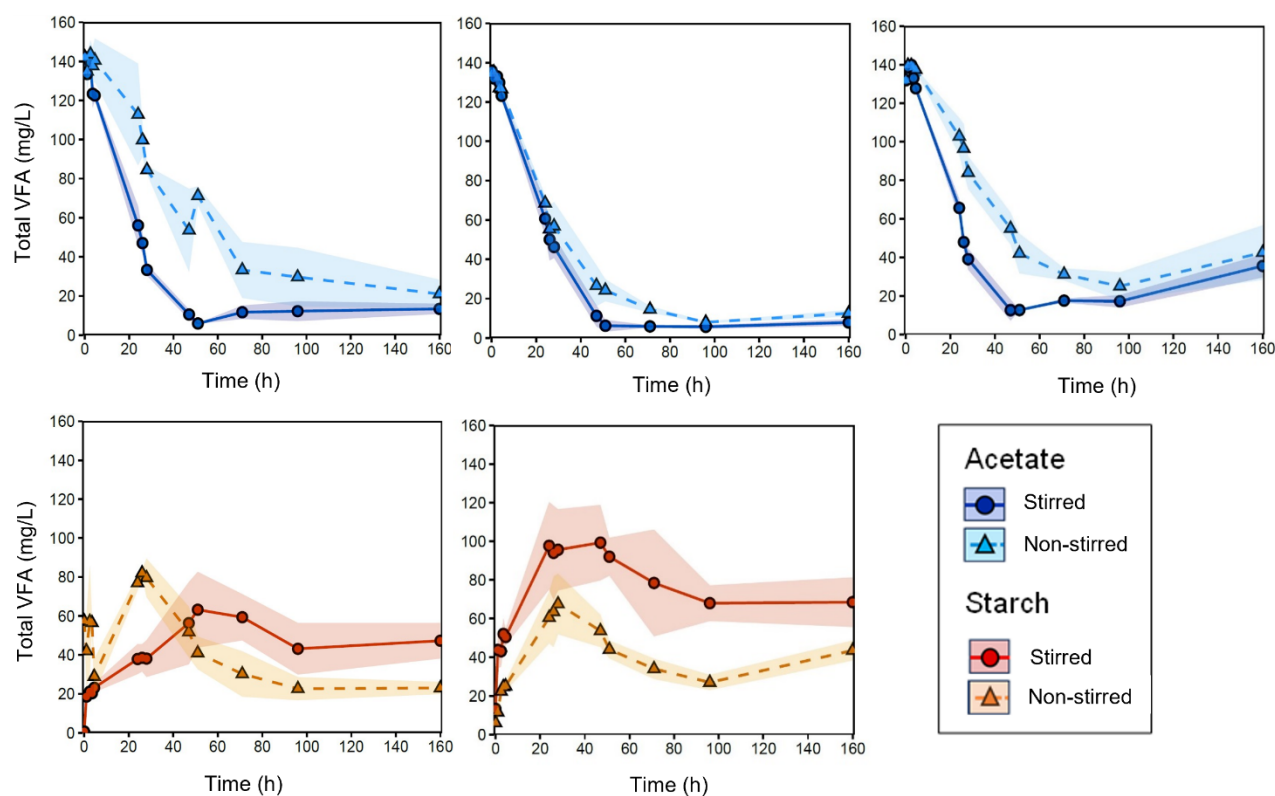


Figure 2. Total VFA production/ consumption during all batch runs for acetate and starch-fed MFCs.

4 Oxygen intrusion measurements

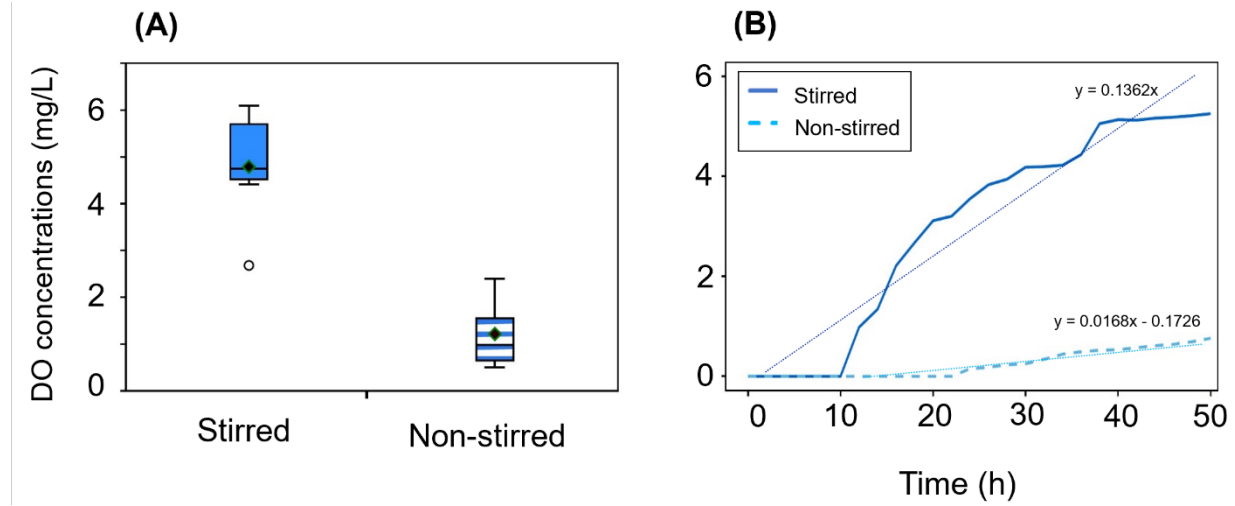
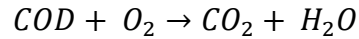


Figure 3. (A) Effluent dissolved oxygen (DO) concentrations for both acetate and starch-fed MFCs under stirred and non-stirred conditions. **(B)** Continuous DO concentrations of a stirred and non-stirred MFC with phosphate buffer.

The amount of COD consumed by incoming oxygen was calculated based on the following calculation taken from Heidrich et al., (2011):



Therefore, 1g of COD is equal to 1/32 mol oxygen (1 mol of O₂ for every 32g of COD).

- Reactor volume 0.33 L
- Length on batch runs: 160 hours
- Acetate MFC influent: 150 mg/LCOD
- Starch MFC influent: 250 mg/LCOD

COD (in g) in MFCs:

$$Acetate = \frac{150 \times 0.33}{1000} = 0.0495 \text{ g}$$

$$Starch = \frac{250 \times 0.33}{1000} = 0.0825 \text{ g}$$

4.1 Stirred MFC calculations

Rate of oxygen intrusion = 0.14 mg/L of oxygen per hour (gradient of line taken from equation, Figure 3B).

$$\text{Rate of } O_2 \text{ entering the system} = \frac{1.4 \times 10^{-4} \text{ g/L/h} \times 0.33 \text{ L}}{32 \text{ g/mol}} = 1.44 \times 10^{-6} \text{ mol/h}$$

$$O_2 \text{ available in batch run} = 1.44 \times 10^{-6} \text{ mol/h} \times 160 \text{ h} = 2.3 \times 10^{-4} \text{ moles}$$

$$COD \text{ available for } O_2 \text{ consumption} = 2.3 \times 10^{-4} \text{ mol} \times 32 = 0.0074 \text{ g COD}$$

Stirred acetate:

$$\text{Percentage of COD available for } O_2 \text{ to consume} = \left(\frac{0.0074 \text{ g}}{0.0495 \text{ g}} \right) \times 100 = \mathbf{15\%}$$

Stirred starch:

$$\text{Percentage of COD available for } O_2 \text{ to consume} = \left(\frac{0.00825 \text{ g}}{0.0495 \text{ g}} \right) \times 100 = \mathbf{9\%}$$

4.2 Non-stirred MFC calculations

Rate of oxygen intrusion = 0.017 mg/L of oxygen per hour (gradient of line taken from equation, Figure 3B).

$$\text{Rate of } O_2 \text{ entering the system} = \frac{1.7 \times 10^{-5} \text{ g/L/h} \times 0.33 \text{ L}}{32 \text{ g/mol}} = 1.75 \times 10^{-7} \text{ mol/h}$$

$$O_2 \text{ available in batch run} = 1.75 \times 10^{-7} \text{ mol/h} \times 160 \text{ h} = 2.8 \times 10^{-5} \text{ moles}$$

$$COD \text{ available for } O_2 \text{ consumption} = 2.8 \times 10^{-5} \text{ mol} \times 32 = 0.0009 \text{ g COD}$$

Non-stirred acetate:

$$\text{Percentage of COD available for } O_2 \text{ to consume} = \left(\frac{0.0009 \text{ g}}{0.0495 \text{ g}} \right) \times 100 = \mathbf{1.8\%}$$

Non-stirred starch:

$$\text{Percentage of COD available for } O_2 \text{ to consume} = \left(\frac{0.0009 \text{ g}}{0.0495 \text{ g}} \right) \times 100 = 1\%$$

5 Alpha diversity plots

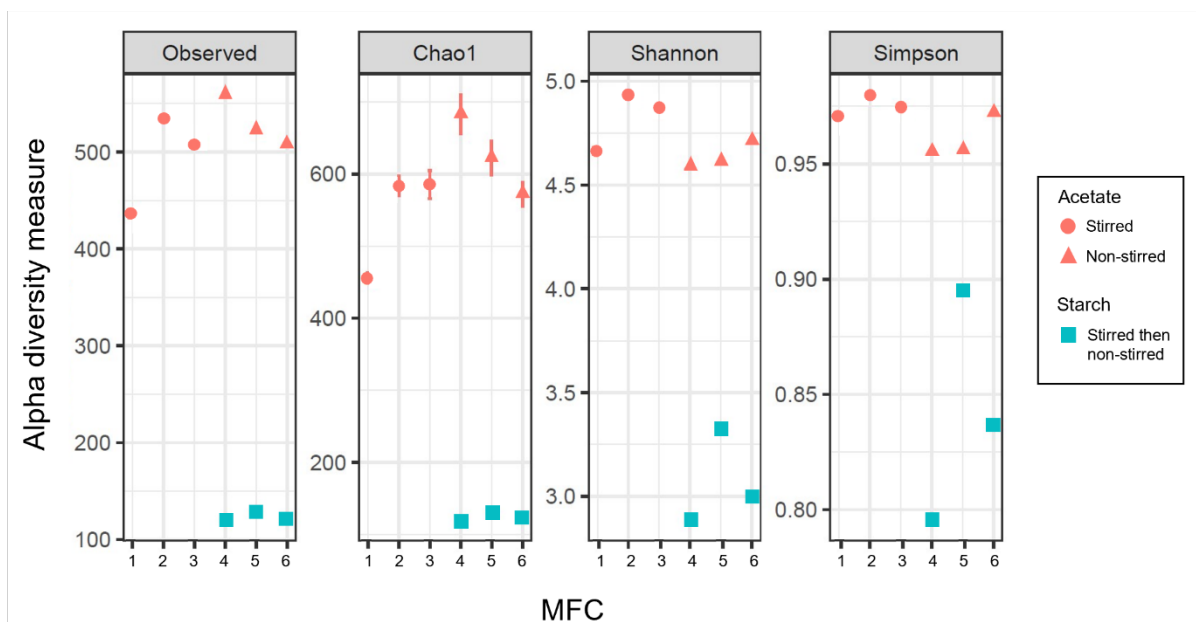


Figure 4. Alpha diversity plots showing the diversity of the anodic microbial communities in the acetate and starch-fed MFCs. As starch reactors underwent both stirring conditions sequentially (stirred then non-stirred), there were only 3 biofilm samples to analyse (denoted by square icon).